

# Администрирование сетевых подсистем

## Настройка DNS-сервера (Лабораторная работа №2)

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13 сентября 2025

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## Цели и задачи работы

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Получение практических навыков установки и настройки DNS-сервера на базе BIND.

## Выполнение лабораторной работы

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## Проверка работы сервера

```
[root@server.manovichkov.net ~]# dig www.yandex.ru

; <<>> DiG 9.18.33 <<>> www.yandex.ru
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 23597
;; flags: qr rd ra; QUERY: 1, ANSWER: 3, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 512
;; QUESTION SECTION:
;www.yandex.ru.                IN      A

;; ANSWER SECTION:
www.yandex.ru.                597     IN      A      77.88.55.88
www.yandex.ru.                597     IN      A      77.88.44.55
www.yandex.ru.                597     IN      A      5.255.255.77

;; Query time: 24 msec
;; SERVER: 10.0.2.3#53(10.0.2.3) (UDP)
;; WHEN: Sat Sep 13 06:22:44 UTC 2025
;; MSG SIZE rcvd: 90

[root@server.manovichkov.net ~]#
```

```
; <<>> DiG 9.18.33 <<>> www.yandex.ru
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 23597
;; flags: qr rd ra; QUERY: 1, ANSWER: 3, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 512
;; QUESTION SECTION:
;www.yandex.ru.                IN      A

;; ANSWER SECTION:
www.yandex.ru.                597     IN      A      77.88.55.88
www.yandex.ru.                597     IN      A      77.88.44.55
www.yandex.ru.                597     IN      A      5.255.255.77

;; Query time: 24 msec
;; SERVER: 10.0.2.3#53(10.0.2.3) (UDP)
;; WHEN: Sat Sep 13 06:22:44 UTC 2025
;; MSG SIZE rcvd: 90

[root@server.manovichkov.net ~]#
```

```
[root@server.manovichkov.net ~]# nmcli connection edit eth0

===| nmcli interactive connection editor |===

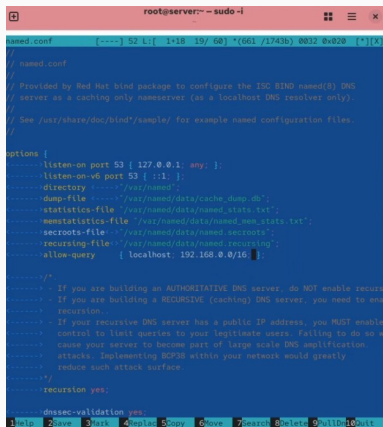
Editing existing '802-3-ethernet' connection: 'eth0'

Type 'help' or '?' for available commands.
Type 'print' to show all the connection properties.
Type 'describe [<setting>.<prop>]' for detailed property description.

You may edit the following settings: connection, 802-3-ethernet (ethernet), 802-
1x, dcb, sriov, ethtool, match, ipv4, ipv6, hostname, link, tc, proxy
nmcli> remove ipv4.dns
nmcli> set ipv4.ignore-auto-dns yes
nmcli> set ipv4.dns 127.0.0.1
nmcli> save
Connection 'eth0' (e292e83a-7750-4087-b4e1-a998fc55c0ea) successfully updated.
nmcli> quit
[root@server.manovichkov.net ~]#
```

Рис. 3: Настройка подключения через nmcli

# Изменения в named.conf



```
root@server:~ - sudo -i
named.conf [----] 52 L: [ 1*18 19/ 60] *(661 /1743b) 0032 0x020 [*][X]

/ named.conf

/ Provided by Red Hat bind package to configure the ISC BIND named(8) DNS
/ server as a caching only nameserver (as a localhost DNS resolver only).

/ See /usr/share/doc/bind*/sample/ for example named configuration files.
/

options {
    -----listen-on port 53 { 127.0.0.1; any; };
    -----listen-on-v6 port 53 { ::1; };
    -----directory "-----"/var/named";
    -----dump-file "-----"/var/named/data/cache_dump.db";
    -----statistics-file "/var/named/data/named.stats.txt";
    -----memstatistics-file "/var/named/data/named_mem_stats.txt";
    -----secroots-file "-----"/var/named/data/named.secroots";
    -----recursing-file "-----"/var/named/data/named.recursing";
    -----allow-query { localhost; 192.168.0.0/16; };

    -----*/
    -----* If you are building an AUTHORITATIVE DNS server, do NOT enable recurs
    -----* If you are building a RECURSIVE (caching) DNS server, you need to ena
    -----* recursion.
    -----* If your recursive DNS server has a public IP address, you MUST enable
    -----* control to limit queries to your legitimate users. Failing to do so w
    -----* cause your server to become part of large scale DNS amplification
    -----* attacks. Implementing BCP38 within your network would greatly
    -----* reduce such attack surface
    -----*/
    -----recursion yes;

    -----dnssec-validation yes;

1)Help 2)Save 3)Mark 4)Replace 5)Copy 6)Move 7)Search 8)Delete 9)Undo 10)Quit
```

Рис. 4: Изменения в named.conf

# Проверка работы DNS

```
[root@server.manovichkov.net ~]# lsof | grep UDP
lsof: WARNING: can't stat() fuse.gvfsd-fuse file system /run/user/1001/gvfs
Output information may be incomplete.
lsof: WARNING: can't stat() fuse.portal file system /run/user/1001/doc
Output information may be incomplete.
avahi-daemon 886      0t0      UDP *:mdns          avahi  12u    IPv4      8190
avahi-daemon 886      0t0      UDP *:mdns          avahi  13u    IPv6      8191
chronyd       970      0t0      UDP localhost:323    chrony  5u     IPv4      8420
chronyd       970      0t0      UDP localhost:323    chrony  6u     IPv6      8421
named         27189    0t0      UDP localhost:domain named  25u    IPv4      67328
named         27189    0t0      UDP localhost:domain named  26u    IPv4      67329
named         27189    0t0      UDP localhost:domain named  31u    IPv6      67332
named         27189    0t0      UDP localhost:domain named  32u    IPv6      67333
named         27189 27190    0t0      UDP localhost:domain named  25u    IPv4      67328
named         27189 27190    0t0      UDP localhost:domain named  26u    IPv4      67329
named         27189 27190    0t0      UDP localhost:domain named  31u    IPv6      67332
named         27189 27190    0t0      UDP localhost:domain named  32u    IPv6      67333
named         27189 27191    0t0      UDP localhost:domain named  25u    IPv4      67328
```

Рис. 5: Проверка работы DNS через lsof

# Создание зоны manovichkov.net

```
root@server:/etc/named - sudo -i

named.conf      [-M--] 37 L: [ 28+31 59/ 60] *(1779/1780b) 0010 0x00A [*][X]
-----> attacks. Implementing BCP38 within your network would greatly
-----> reduce such attack surface.
----->*/
----->recursion yes;

----->dnssec-validation yes;

----->managed-keys-directory "/var/named/dynamic";
----->geoup-directory "/usr/share/GeoIP";

----->pid-file "/run/named/named.pid";
----->session-keyfile "/run/named/session.key";

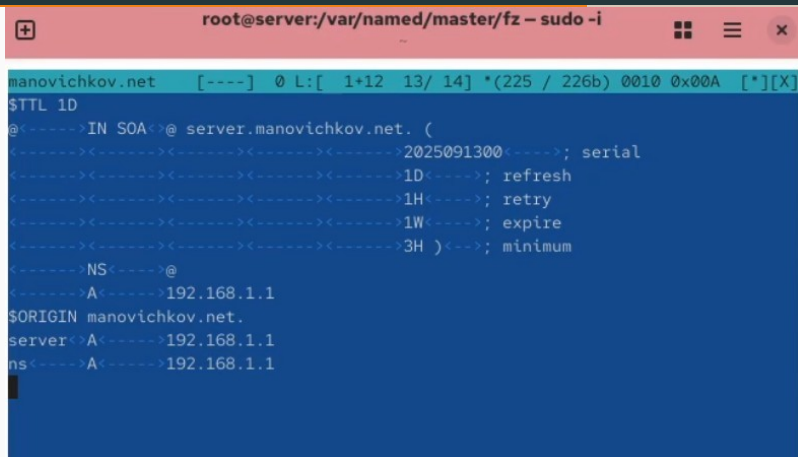
----->/* https://fedoraproject.org/wiki/Changes/CryptoPolicy */
----->include "/etc/crypto-policies/back-ends/bind.config";
};

logging {
    channel default_debug {
        file "data/named.run";
        severity dynamic;
    };
};

zone "." IN {
    ----->type hint;
    ----->file "named.ca";
};

include "/etc/named.rfc1912.zones";
include "/etc/named.root.key";
include "/etc/named/manovichkov.net";

1Help 2Save 3Mark 4Replac 5Copy 6Move 7Search 8Delete 9PullDn10Quit
```



```
root@server:/var/named/master/fz - sudo -i
manovichkov.net  [----] 0 L:[ 1+12 13/ 14] *(225 / 226b) 0010 0x00A [*][X]
$TTL 1D
@<----->IN SOA<@ server.manovichkov.net. (
<-----><-----><-----><-----><----->2025091300<----->; serial
<-----><-----><-----><-----><----->1D<----->; refresh
<-----><-----><-----><-----><----->1H<----->; retry
<-----><-----><-----><-----><----->1W<----->; expire
<-----><-----><-----><-----><----->3H )<-->; minimum
<----->NS<----->@
<----->A<----->192.168.1.1
$ORIGIN manovichkov.net.
server<A<----->192.168.1.1
ns<----->A<----->192.168.1.1
```

Рис. 7: Файл прямой зоны manovichkov.net

```
root@server:/var/named/master/rz - sudo -i
192.168.1 [----] @ L: [ 1+13 14/ 15] *(279 / 280b) 0010 0x00A [*](X)
$TTL 1D
@ IN SOA @ server.manovichkov.net. (
    2025091300 ; serial
    1D ; refresh
    1H ; retry
    1W ; expire
    3H ; minimum
)
@ IN NS @
@ IN A 192.168.1.1
@ IN PTR server.manovichkov.net.
$ORIGIN 1.168.192.in-addr.arpa.
1 IN PTR server.manovichkov.net.
1 IN PTR ns.manovichkov.net.
```

Рис. 8: Файл обратной зоны 192.168.1

```
[root@server.manovichkov.net rz]# chown -R named:named /etc/named
[root@server.manovichkov.net rz]# chown -R named:named /var/named
[root@server.manovichkov.net rz]# restorecon -vR /etc
Relabeled /etc/lvm/devices/system.devices from system_u:object_r:lvm_metadata_t:s0 to system_u:object_r:lvm_etc_t:s0
Relabeled /etc/lvm/devices/backup/system.devices-20250913.061712.0005 from system_u:object_r:lvm_metadata_t:s0 to system_u:object_r:lvm_etc_t:s0
Relabeled /etc/NetworkManager/system-connections/eth1.nmconnection from unconfined_u:object_r:user_tmp_t:s0 to unconfined_u:object_r:NetworkManager_etc_rw_t:s0
[root@server.manovichkov.net rz]# restorecon -vR /var/named
[root@server.manovichkov.net rz]# getsebool -a | grep named
named_tcp_bind_http_port --> off
named_write_master_zones --> on
[root@server.manovichkov.net rz]#
```

Рис. 9: Настройка прав и SELinux

# Конфигурация зон

```
root@server:/var/named/master/fz -- sudo -i

manovichkov.net  [-M--] 39 L:[ 1+18 19/ 28] *(567 / 705b) 0034 0x022  [*][X]
// named.rfc1912.zones:
//
// Provided by Red Hat caching-nameserver package.
//
// ISC BIND named zone configuration for zones recommended by
// RFC 1912 section 4.1 : localhost TLDs and address zones
// and https://tools.ietf.org/html/rfc6303
// (c)2007 R W Franks
//
// See /usr/share/doc/bind*/sample/ for example named configuration files.
//
// Note: empty-zones-enable yes; option is default.
// If private ranges should be forwarded, add
// disable-empty-zone "."; into options
//
zone "manovichkov.net" IN {
----->type master;
----->file "master/fz/manovichkov.net";
----->allow-update { none; };
};

zone "1.168.192.in-addr.arpa" IN {
----->type master;
----->file "master/rz/192.168.1";
----->allow-update { none; };
};

1Help 2Save 3Mark 4Replac 5Copy 6Move 7Search 8Delete 9PullDn 10Quit
```

```
[root@server.manovichkov.net fz]# dig ns.manovichkov.net

; <<>> DiG 9.18.33 <<>> ns.manovichkov.net
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 26255
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
; COOKIE: 88b461edec54836b0100000068c515981cd384f640eac686 (good)
;; QUESTION SECTION:
ns.manovichkov.net.                IN      A

;; ANSWER SECTION:
ns.manovichkov.net.                86400   IN      A      192.168.1.1

;; Query time: 0 msec
;; SERVER: 127.0.0.1#53(127.0.0.1) (UDP)
;; WHEN: Sat Sep 13 06:56:24 UTC 2025
;; MSG SIZE rcvd: 91

[root@server.manovichkov.net fz]#
```

## Проверка работы зоны через host

```
[root@server.manovichkov.net fz]# host -l manovichkov.net
manovichkov.net name server manovichkov.net.
manovichkov.net has address 192.168.1.1
ns.manovichkov.net has address 192.168.1.1
server.manovichkov.net has address 192.168.1.1
[root@server.manovichkov.net fz]# host -a manovichkov.net
Trying "manovichkov.net"
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 8647
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 3, AUTHORITY: 0, ADDITIONAL: 0

;; QUESTION SECTION:
;manovichkov.net.          IN      ANY

;; ANSWER SECTION:
manovichkov.net.          86400   IN      SOA      manovichkov.net. server.manovichkov.net. 2025091300 86400 3600 604
800 10800
manovichkov.net.          86400   IN      NS       manovichkov.net.
manovichkov.net.          86400   IN      A        192.168.1.1

Received 106 bytes from 127.0.0.1#53 in 0 ms
[root@server.manovichkov.net fz]# host -t A manovichkov.net
manovichkov.net has address 192.168.1.1
[root@server.manovichkov.net fz]# host -t PTR 192.168.1.1
1.1.168.192.in-addr.arpa domain name pointer ns.manovichkov.net.
1.1.168.192.in-addr.arpa domain name pointer server.manovichkov.net.
```

Рис. 12:Проверка зоны через host

## Выводы по проделанной работе

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В ходе лабораторной работы был установлен и настроен DNS-сервер на основе BIND. Реализованы прямая и обратная зоны, проведена проверка их работы через `dig` и `host`. Настроен кэширующий сервер и автоматизация конфигурации с помощью Vagrant.